

RANGAMATI SCIENCE AND TECHNOLOGY UNIVERSITY

Department of Computer Science and Engineering

4th Year 1st Semester BSc (Engg.) 1st Midterm Examination - 2023

Session: 2019-2020

Course Code: CSE 4102; Course Title: Computer Networking

Duration: 1 Hour;

Full Marks: 15

Answer all questions

1. Briefly explain the layers of the TCP/IP model. 2.5
2. a) Describe the physical layer, highlighting its key components and common issues associated with it. 2.5
b) How are issues in the physical layer addressed by the upper layer? Illustrate your explanation with a suitable diagram. 2.5
3. Write notes on the following: 05
 - i) MAC Address
 - ii) IP Addressing System
4. Explain the statement: "IP addresses belong to networks, not to the devices attached to those networks." 2.5

Rangamati Science and Technology University

Department of Computer Science and Engineering

4thYear 1st Semester Mid -1 Examination-2025

Session: 2019-2020

Course Title: Computer Graphics Course Code: CSE 4104

Time- 1 hour

Total Marks-15

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|----------|----|---|----------|
| 1 | a. | Define computer graphics. Mention the basic components of computer graphics. | 3 |
| | b. | Shortly describe the RGB and CMY color model. | 2 |
| 2 | a. | Describe the working principle of cathode ray tube- CRT with its fundamental units. | 3 |
| | b. | Point out the applications of computer graphics. | 2 |
| 3 | a. | Define and plot the successive generations of the C curve & Koch curve at level 4 of a straight line. | 3 |
| | b. | Show the different types of computer graphics. | 2 |

Rangamati Science and Technology University
Department of Computer Science and Engineering
4th Year 1st Semester B.Sc. Engg. 1st mid Term Examination-2023
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Course Title: Natural Language Processing

Course Code: CSE 5105

Time -1 hrs

Total Marks-15

1. [3 Marks]

Define and explain the core components of Natural Language Processing (NLP).

Use the sentence below to demonstrate any two components:

> "The quick brown fox jumps over the lazy dog."

2. [3 Marks]

Differentiate between high-resource and low-resource languages in NLP.

Provide one example of each and discuss the key challenges in low-resource NLP processing.

3. [3 Marks]

Why have Machine Learning (ML), Deep Learning (DL), and Transformer-based models been successively adopted in NLP?

Briefly analyze one key improvement each offers over the previous method.

4. [3 Marks]

How does Machine Learning improve over rule-based systems in Natural Language Processing?

Justify with an example such as sentiment analysis or POS tagging.

5. [3 Marks]

You are given a dataset of multilingual and code-mixed text (e.g., Bangla-English).

a) Identify at least three key challenges in processing this type of text in NLP.

b) Justify your model choice (e.g., multilingual transformers, hybrid models).



Rangamati Science and Technology University

Department of Computer Science and Engineering

4th Year 1st Semester, Mid-term 1, Examination-2023

Subject: Financial, Cost and Managerial Accounting

Couse Code: CSE 4106

Full Marks: 15

Time: 1 Hour

Answer the Following Questions:

1. "Bookkeeping and accounting are the same." Do you agree? Explain. [3]
2. On April 1, Richard established Richard's Travel Agency. The following transactions [8+4]
were completed during the month.
 1. Invested \$20,000 cash to start the agency.
 2. Paid \$600 cash for April office rent.
 3. Purchased equipment for \$3,000 cash.
 4. Incurred \$700 of advertising costs in the Chicago Tribune, on account.
 5. Paid \$900 cash for office supplies.
 6. Performed services worth \$10,000: \$3,000 cash is received from customers, and the balance of \$7,000 is billed to customers on account.
 7. Withdrew \$1000 cash for personal use.
 8. Paid Chicago Tribune \$500 of the amount due in transaction (4).
 9. Paid employees' salaries \$2,500.
 10. Received \$4,000 in cash from customers who have previously been billed in transaction (6).

Instructions

(a) Prepare a tabular summary of the transactions.

(b) Prepare the income statement and owner's equity statement for the month of April.